

Run-to-failure data set of ball bearings subjected to time-varying load and speed conditions

Osarenren Kennedy Aimiyekagbon
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1 Introduction

This data set consist of experimental data collected during 17 run-to-failure experiments on ball bearings subjected to time-varying load and speed conditions. No defect was initiated in the bearings before the experiments. The 61806-2RS test bearing characteristics are listed in table 1 and the respective dimensions are depicted in Figure 1.

Table 1: Characteristics of tested 61806-2RS bearing

Attribute	Value
Inner diameter d	30 mm
Outer diameter D	42 mm
Width B	7 mm
Rolling circle diameter D_T	35.5 mm
Rolling element diameter D_W	3.18 mm
Number of balls z	19
Seal	NBR+08
Basic static load rating C_0	3.15 kN
Basic dynamic load rating C	4.00 kN

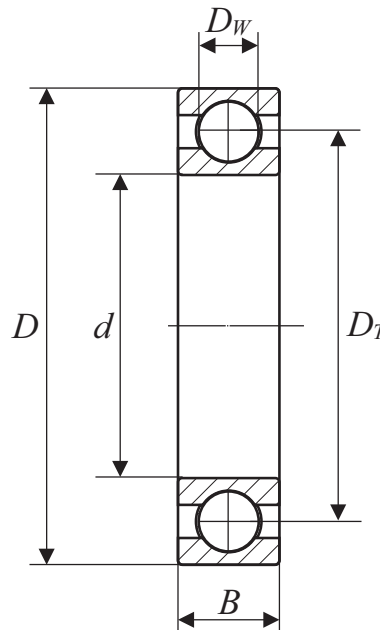


Figure 1: Ball bearing dimensions

The test rig consists of the housed test bearing (2), which is mounted on the shaft, as seen in Figure 2(a). The shaft is supported by two spherical roller bearings (6) and driven by the motor (1). The total load on the test bearing is a combination of a static pre-load and a superimposed dynamic load. The static pre-load is exerted on the bearing via a lever structure (4) and the dynamic load by means of an electrodynamic shaker (5). A detailed description of the bearing test rig and experimental procedure can be found in our corresponding paper to follow.

The operating conditions are measured synchronously with vibration and temperature to monitor the test bearing's health. Three one-directional accelerometers (4) measure the vibration of the bearing via its housing. However, two (A and C) of the three accelerometer measurements, as illustrated in Figure 2(b) are provided in this data set, because the accelerometer is farther away from the test bearing. Furthermore, the ambient temperature and bearing temperature measurements are also provided. The bearing temperature is measured at the positions (T1 and T2) depicted in Figure 2(b). The ambient temperature is measured in close proximity to the test bearing and support bearings.

During an experiment the operating conditions, that is the shaft rotating speed and the amplitudes of the dynamic load, are randomly drawn from a stationary uniform distribution within a predefined interval between measurements. For experiments B01 through B06, the dynamic load is sinusoidal with a constant frequency of 2 Hz and for experiments B07 through B17, the dynamic load is Gaussian white noise with a maximum excitation frequency of 200 Hz. The predefined range of values per experiment is in Table 2. For some experiments, only one operating condition, such as the amplitude of the static load is from an interval.

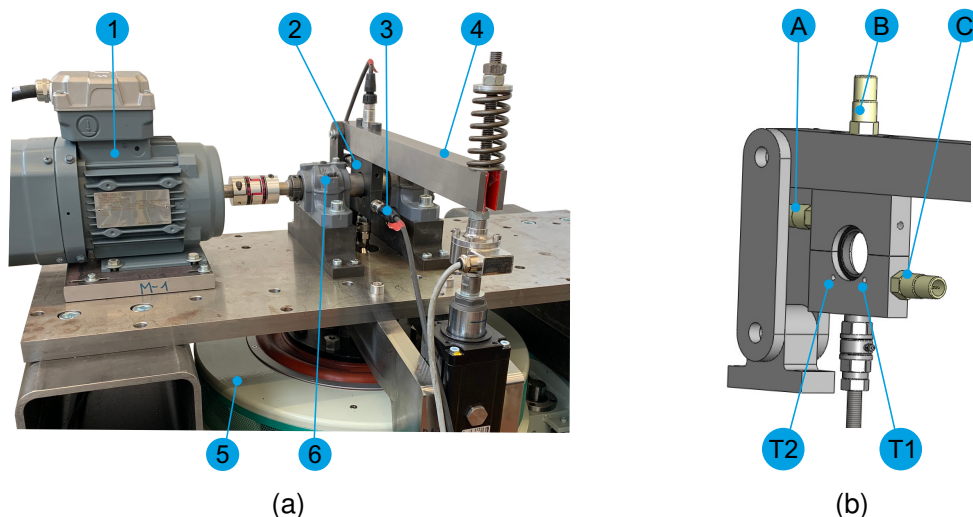


Figure 2: (a) Ball bearing test-rig with the following components: (1) motor, (2) bearing housing, (3) accelerometers, (4) lever structure, (5) electrodynamic shaker (6) support bearing housing.
(b) Accelerometer and temperature placement on the test bearing housing without the shaft

Table 2: Set range of operating conditions for experiments

Experiment	Rotating Speed [rpm]	Static Load Amplitude [N]	Dynamic Load Amplitude [N]	Dynamic Load Excitation
B01 - B03	[2400, 3000]	[3300, 3800]	[500, 1400]	Sinusoidal
B04 - B05	[1500, 3000]	[2500, 3800]	[500, 1500]	Sinusoidal
B06	[1500, 3600]	[3300, 3800]	[750, 1700]	Sinusoidal
B07	[1500, 3000]	[3250, 4000]	[250, 750]	Gaussian noise
B08	[1500, 3000]	[3250, 4000]	[500, 1000]	Gaussian noise
B09	[1500, 3000]	[2500, 3800]	[750, 1000]	Gaussian noise
B10	[1500, 2700]	[2000, 3250]	[750, 1500]	Gaussian noise
B11 - B13	[1500, 2700]	3000	1000	Gaussian noise
B14 - B15	2700	2500	[700, 1500]	Gaussian noise
B16	2700	2500	[1000, 1500]	Gaussian noise
B17	2700	[1500, 2500]	1500	Gaussian noise

A bearing fails and consequently an experiment ends, when at least one of two predetermined failure threshold criteria is exceeded. However, for experiment B01 no criteria was pre-set and for B05 the experiment was deliberately interrupted. The threshold criteria is based on the bearing housing vibration $T_{vibration}$ and temperature $T_{temperature}$ measurements. For the provided experiments,

- $T_{vibration}$ lies within [6, 10] g and
- $T_{temperature}$ is 110 °C.

2 Measurement and Data Details

All control signals and measurements were made with the MATLAB[®] data acquisition (DAQ) toolbox and National Instruments (NI[™]) cDAQ-9189. Measurements were acquired at a sampling duration of 1.6 s and a measurement interval of approximately 12 s. The temperature and accelerometer measurement chains are laid out in Figures 3 and 4 respectively. Temperature signals are measured with Measurement Computing (MC) 745691-02 Pt100 resistance thermometers at a sampling rate of 10 Hz .

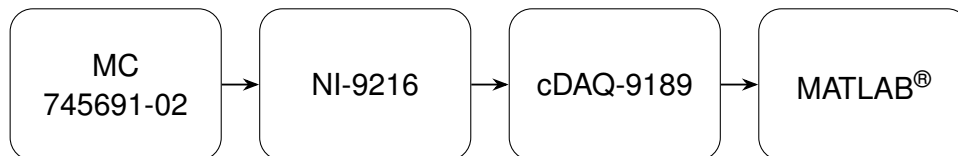


Figure 3: Temperature measurement chain

Thereafter, the signals are digitalised with the NI-9216 and passed on to the DAQ-Hardware and Software. Vibration signals are measured with PCB[®] Piezotronics 603C01 accelerometers, pre-amplified with the MMF M208B signal conditioner and digitalised with the NI-9223 and acquired with the following hard- and software. To facilitate high frequency analysis, vibration data were sampled at 128 kHz for experiments till B09 and due to data storage issues at 64 kHz for experiments from B010. This lower sampling frequency is theoretically sufficient for analysis in the frequency range of interest up to 32 kHz.



Figure 4: Accelerometer measurement chain

2.1 Sensor Specifications

The specifications for the accelerometer and temperature sensor are provided below.

Accelerometer Specifications

- Manufacturer: PCB[®] Piezotronics
- Model: 603C01
- Sensitivity ($\pm 10\%$): 100 mV/g
- Measurement range: ± 50 g
- Frequency range (3 dB): [0.5, 10 000] Hz

- Resonant Frequency: 25 kHz
- Non-Linearity: $\pm 1\%$

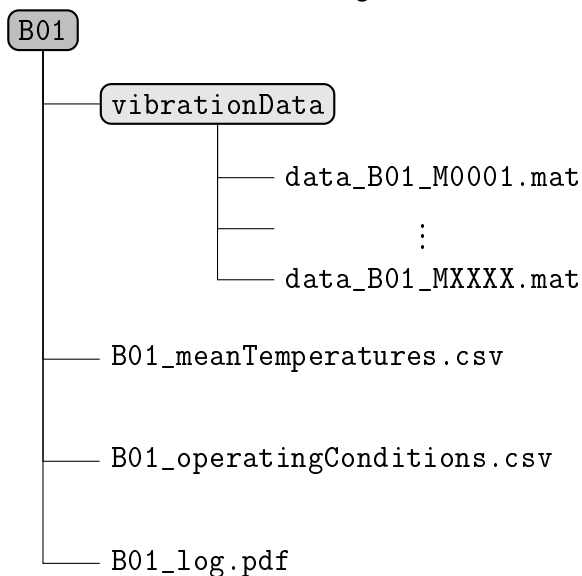
Temperature Sensor Specifications

- Manufacturer: Measurement Computing (MC)
- Model: 745691-02
- Type: Platinum
- Resistance: $100\ \Omega$ at $0\ ^\circ\text{C}$
- Measurement range: $[-50, 204]\ ^\circ\text{C}$
- Accuracy: $\pm(0.3 + 0.005|t|)\ ^\circ\text{C}$, where $|t|$ is the absolute temperature value in $^\circ\text{C}$

2.2 Data set overview

Each experiment folder contains a sub-folder and four files. A tree of an exemplary experiment folder is in the diagram below.

Figure 5: Overview of the experiment B01 folder



The **vibrationData** folder contains the raw vibration signals from directions A and C, as depicted in figure 2(b) and described in Chapter 1. The naming convention for the vibration signal is **data_B01_MXXXX.mat**. Where B01 uniquely identifies an experiment and MXXXX is an incremental measurement identifier. When loaded, each vibration signal file contains the variables `accHorizRear_A`, `accHorizFrontal_C` and `measTime`, which represent the vibration signals from directions A and C and the corresponding sample time, respectively. The signals are given in Voltage (V) and the conversion factor(s) can be found in the corresponding log file. A possible code

for loading a measurement file in Python is given below, which generates a dictionary with the aforementioned variables.

```
1 import scipy.io
2 folderpath = 'dataSetPath/B01/vibrationData/'
3 loadData = scipy.io.loadmat(folderpath+'data_B01_M0001.mat')
```

The **B01_meanTemperatures.csv** file is a table with the columns `dateTime`, `Mean Abs. Temp. T1 / °C`, `Mean Abs. Temp. T2 / °C` and `Mean Amb. Temp. / °C`. The temperatures are given in degree Celsius (°C) and are described as follows:

1. `dateTime` corresponds to the measurement date and time,
2. `Mean Abs. Temp. T1 / °C` corresponds to the measured absolute temperature at the position T1, as denoted in Figure 2(b),
3. `Mean Abs. Temp. T2 / °C` corresponds to the measured absolute temperature at the position T2, as denoted in Figure 2(b),
4. `Mean Amb. Temp. / °C` corresponds to the ambient temperature.

The **B01_meanTemperatures.csv** file and subsequent `.csv` files can be read in Python with the following code.

```
1 import pandas as pd
2 folderpath = 'dataSetPath/B01/'
3 tempDataFrame = pd.read_csv(folderpath+'B01_meanTemperatures.csv')
```

The **B01_operatingConditions.csv** file contains the operating conditions of an experiment. The table columns are `dateTime`, `setDynLoad / N`, `peak_dynLoad / N`, `setStatLoad / N`, `meanAbs_statLoad / N`, `setSpeed / rpm`, `meanAbs_speed / rpm`. The loads are given in Newton (N) and the speeds in revolutions per minute (rpm). The columns are described as follows:

1. `dateTime` is the measurement date and time,
2. `setDynLoad / N` is the set amplitude of the sinusoidal dynamic load or the maximum allowable excitation of the Gaussian noise,
3. `peak_dynLoad / N` is the maximum value of the measured dynamic load,
4. `setStatLoad / N` is the set amplitude of the static pre-load,
5. `meanAbs_statLoad / N` is the mean absolute value of the measured static pre-load,
6. `setSpeed / rpm` is the set shaft rotating speed,
7. `meanAbs_speed / rpm` is the mean absolute value of the measured shaft rotating speed.

B01_extractedFeatures.csv contains a subset of the features extracted from the raw vibration data. This subset was utilised in our paper to follow.

B01_log.pdf is a experiment-specific log file.

Due to the size limit of the repository. The data set is split into several data sets. An overview of the complete data set is given in Table 3. It contains the approximate duration of an experiment in hours, the total number of files, the relevant threshold criterion for the end of an experiment and the defect type discovered after dismantling the test bearing.

Table 3: Data Set Characteristics

Experiment	Approx. Duration	No. of Files	Threshold Criterion	Defect
B01	2 h	377	-	IR, OR, B
B02	6 h	1116	$T_{vibration}$	IR
B03	3 h	614	$T_{temperature}$	IR
B04	6 h	1114	$T_{vibration}$	IR, B
B05	3 h	572	-	B
B06	67 h	12965	$T_{vibration}$	IR, B
B07	37 h	6393	$T_{temperature}$	IR
B08	11 h	1827	$T_{temperature}$	IR, B
B09	13 h	1813	$T_{vibration}$	OR, B
B10	19 h	3224	$T_{vibration}$	OR, B
B11	8 h	1953	$T_{vibration}$	IR, OR, B
B12	3 h	767	$T_{vibration}$	IR, B

Where $T_{vibration}$ and $T_{temperature}$ denote the vibration and threshold value, respectively. IR denotes inner race defect(s), OR denotes outer race defect(s) and B denotes rolling element defect(s).

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Table 3: Data Set Characteristics ... (continued)

Experiment	Approx. Duration	No. of Files	Threshold Criterion	Defect
B13	66 h	19417	$T_{vibration}$	IR, B
B14	41 h	12317	$T_{vibration}$	B
B15	76 h	22567	$T_{temperature}$	IR, B
B16	20 h	5891	-	IR, OR, B
B17	17 h	2733	$T_{vibration}$	B

Where $T_{vibration}$ and $T_{temperature}$ denote the vibration and threshold value, respectively. IR denotes inner race defect(s), OR denotes outer race defect(s) and B denotes rolling element defect(s).